Yuxiao Li

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Education Background

Nanyang Technological University Sep 2023-Jun 2027
Bachelor of Science (Expected in Jun 2027)

- Double Major: Mathematics and Computer Science
- GPA: 4.54/5.0

Internship Experience

KLASS Engineering and Solutions Pte Ltd | Singapore Jan 2026-May 2026
AI & Software Engineering Intern

- Designed and delivered a solo LangGraph-based reimbursement verification agent to process receipts, invoices, and bank statements, implemented a multi-stage state pipeline featuring fuzzy matching for merchant deduplication and a strict-to-lenient review fallback, and reduced end-to-end latency from 2 minutes to 30 seconds through asynchronous parallel parsing and task-complexity-based model routing
- Migrated retriever and data-prep microservices from OPEA to native FastAPI, decoupled the shared RAG library by injecting configurations to enable cross-service reuse, and deployed containerized services using Docker and Gunicorn with a multi-tier storage system utilizing PostgreSQL, S3, and Milvus
- Audited end-to-end behaviors of an agentic document system to identify several critical system bottlenecks, engineered a query rewriter to resolve date-filtering bugs by parsing relative dates into explicit ranges before the tool layer, and integrated string-similarity merchant deduplication fixes into production

SRIT Information | Ningbo, China Jul 2025-Aug 2025
Intern

- Engineered a Model Context Protocol (MCP) server backend for a cancer diagnosis support system
- Incorporated a Retrieval-Augmented Generation (RAG) model to query local diagnosis documentation, and improved answer relevance

Project Experience

Final Year Paper: Interpretable Irregular Time-Series Modelling Oct 2026-Present
Mentored by Lecturer Siyuan Liu, Nanyang Technological University

- Reviewed irregular ICU time-series literature, analyzed Neural ODE computational efficiency against time-blind baselines (e.g., GraFITi-C vs. Physiome-ODE), and established GRU-D with decay/mask mechanisms as the initial evaluation baseline
- Designed a Temporal Concept Bottleneck Model (CBM) framework for ICU monitoring, mapped mimic iv dataset to instantaneous concepts, and proposed an ODE-based propagation space to model dynamic clinical states over irregular intervals

URECA: Agentic LLM System for Virtual Lab Assessment Jun 2026
Undergraduate Researcher, mentored by Prof. Kang Hao Cheong, NTU

- Developed a deterministic Python parser to reconstruct student action timelines from database session logs, resolved obscure internal machine identifiers into human-readable scientific terms using embedded templates, and bypassed LLM dependency in the reconstruction stage to ensure complete auditability
- Engineered an LLM-based diagnosis engine to identify procedural errors, authored a comprehensive technical research paper draft, and presented research milestones in bi-weekly academic seminars

URECA: Fake News Identification Aug 2024-Jun 2025
Undergraduate Researcher, mentored by Prof. Kang Hao Cheong, NTU

- Designed a three-stream hybrid neural network in PyTorch to identify fake news, integrated Word-Level KAN, BERT-BiLSTM, and CNN-MHSA branches to capture semantic references, sequential context, and local attention patterns, and fused the multi-stream feature vectors via concatenation for final classification

- Pioneered an intrinsically explainable Word-Level KAN branch to resolve interpretability limitations, mapped input words to continuous semantic profiles by calculating similarity against 32 credibility anchor words (including truth, lie, and fact), and leveraged KAN's learnable edge functions to trace inspectable transformations and word-level contributions
- Evaluated the hybrid architecture on a composite dataset to achieve a 98.46% classification accuracy, authored a technical research report detailing attention heatmaps, branch contributions, and linguistic style limitations, and presented research milestones in bi-weekly progress reviews with Dr. Hu Shiyu and the research group

Course Project Experience

SC3020 Database Systems Principles Oct 2026-Nov 2026

Project 1

- Developed the database execution driver to orchestrate the query pipeline, implemented a storage-layer logical deletion mechanism using tombstones, and preserved the physical address stability of active records to prevent index pointer invalidation
- Optimized disk I/O operations by sorting record addresses to group storage-layer operations by block ID, eliminated redundant block access, and implemented a linear scan baseline for performance comparison

Project 2

- Developed a query-plan comparison algorithm, aligned execution paths using Longest Common Subsequence (LCS) dynamic programming, and resolved table-matching ambiguities in self-joins by calculating the insertion/deletion (Indel) edit distance of node routes
- Parsed and normalized SQL predicates using sqlglot, and formulated a normalized difference rate metric (ranging from 0 to 1) based on heuristic severities to score plan discrepancies

SC4000 Machine Learning: Writing Quality Prediction Sep 2026-Oct 2026

- Preprocessed a sequential keystroke dataset of approximately 2,500 essays by removing idle periods and capping abnormal event durations, reconstructed the structural layout and character sequences of masked texts from sequential edit operations, and transformed raw event logs into structured, model-ready inputs
- Extracted multi-dimensional behavioral features focusing on writing pacing and structural patterns, applied character-level Latent Dirichlet Allocation (LDA) to capture morphological distributions despite text masking, and utilized t-SNE to generate low-dimensional representations of complex temporal interactions
- Contributed to a multi-level ensemble pipeline stacking out-of-fold predictions from LightGBM, XGBoost, and CatBoost via Ridge regression, implemented 5-fold cross-validation and early stopping to mitigate overfitting risks under limited samples, and achieved a final RMSE of 0.5954 (a 0.0114 improvement over the LightGBM baseline

Individual Project: Course Schedule Website Jan 2025-May 2025

- Architected a dynamic scheduling application with Next.js, TypeScript, and Tailwind CSS to address inefficient manual course planning
- Collaborated with a two-person backend team to integrate their schedule generation API and partnered with another four-person team by enabling calendar API exports, contributing to a platform that attracted over 2,000 unique visitors in its first month

Technical Skills & Languages

Languages: Mandarin (native), English (proficient)

Programming: `Python`, `Java`, `C++`, `R`, `LaTeX`, `JavaScript`, `HTML/CSS`, `Elisp`

AI/ML: `PyTorch`, `TensorFlow`, `Scikit-learn`, `Pandas`, `OpenCV`

Developer Tools: `Emacs`, `Git`, `Docker`, `JetBrains IDEs`

Platforms: Linux (Debian, Fedora, Ubuntu), macOS